PECO311C : Machine Learning: Concepts and Applications		
Teaching Scheme	**Examination Scheme**	
Lectures: 3 Hrs./Week	**CIA:**	**40 Marks**
	ESE:	**60 Marks**
Credits: 3	**Total:**	**100 Marks**
Prerequisite Course: Problem solving using Python, Application development using Python		

Course Objectives

1. To impart foundational knowledge of machine learning concepts, mathematical principles, and supervised/unsupervised learning algorithms relevant to industrial applications.
2. To develop the ability to analyze and preprocess real-world datasets using appropriate techniques for data exploration, visualization, and feature engineering.
3. To equip students with practical skills to design, implement, and deploy machine learning solutions using modern frameworks such as TensorFlow, PyTorch, and Scikit-learn
4. To cultivate critical thinking for evaluating ML models in terms of performance, fairness, and ethical considerations, along with understanding their societal and environmental impacts
5. To encourage innovation and lifelong learning by exploring cutting-edge trends in machine learning, including generative AI, graph neural networks, and edge AI technologies

Course Outcomes (COs):

After successful completion of the course, student will be able to

	Course Outcome (s)	Bloom's Taxonomy	
		Level	Descriptor
CO1	Apply fundamental and advanced machine learning algorithms to solve practical and industrial problems	3	Apply
CO2	Analyze real-world datasets using appropriate ML techniques to derive insights and support decision-making in industrial contexts	3	Apply
CO3	Design and implement machine learning-based solutions using modern tools and frameworks suited for industry deployment	3	Apply
CO4	Evaluate ML models for accuracy, efficiency, and ethical considerations and reflect on their societal and environmental impact	3	Apply
CO5	Continuously explore emerging trends and technologies in applied machine learning to innovate and upgrade solutions in dynamic industry environments	3	Apply

Mapping of Course Outcomes to Program Outcomes (POs) & Program Specific Outcomes (PSOs):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	3	-	-	-	-	-	-	-	-	3	-	-
CO2	3	3	3	2	2	-	-	1	-	-	-	3	-	-
CO3	3	3	3	3	3	-	-	-	-	-	-	3	2	-
CO4	2	-	2	-	2	2	2	3	-	2	2	2	-	-
CO5	2	2	3	2	3	-	-	-	-	-	-	2	3	3

Course Contents			
Unit-I	**FUNDAMENTALS OF APPLIED MACHINE LEARNING**	**No. of Hours**	**COs**
	Introduction to Machine Learning Overview of machine learning concepts, types (supervised, unsupervised, reinforcement), and real-world applications in industry. Mathematical Foundations Basics of linear algebra (vectors, matrices), calculus (derivatives, gradients), and probability theory essential for understanding ML algorithms. Data Preprocessing and Exploration Techniques for data cleaning, normalization, handling missing values, and exploratory data analysis. Introduction to Supervised Learning Algorithms Linear Regression, Logistic Regression, Decision Trees, and Support Vector Machines, including their mathematical underpinnings and applications. Model Evaluation and Validation Techniques Understanding overfitting, underfitting, cross-validation methods.	9	CO1
Unit-II	**DATA ANALYSIS AND MACHINE LEARNING TECHNIQUES FOR INDUSTRIAL APPLICATIONS**	**No. of Hour s**	**COs**
	Exploratory Data Analysis (EDA) Techniques for summarizing and visualizing datasets to uncover patterns, spot anomalies, and test hypotheses. Feature Engineering and Selection Methods for creating new features, selecting relevant features, and reducing dimensionality to improve model performance. Supervised Learning Algorithms Decision Trees, Random Forests, and Gradient Boosting Machines, including their applications in industry. Unsupervised Learning Techniques K-Means, Hierarchical Clustering, and DBSCAN, and their use cases in industrial scenarios.	9	CO2
Unit-III	**DESIGNING AND IMPLEMENTING ML SOLUTIONS WITH MODERN TOOLS**	**No. Of Hours**	**COs**
	Introduction to ML Frameworks TensorFlow, PyTorch, and Scikit-learn. Understanding their architecture, functionalities, and suitability for various industrial applications. Model Development and Training Hands-on experience in building, training, and validating machine learning models using real-world datasets. Emphasis on best practices for model development, including data preprocessing, feature engineering, and hyperparameter tuning. Deployment Strategies model serialization, Flask- RESTful APIs, containerization using Docker, and deployment on cloud platforms like AWS and Azure.	9	CO3

Unit-IV	EVALUATION METRICS, ETHICAL CONSIDERATIONS, AND SOCIETAL IMPACT OF ML MODELS	No. of Hours	COs
	Model Evaluation Metrics Understanding key performance metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix to assess model effectiveness. Bias and Fairness in Machine Learning Identifying and mitigating biases in ML models	9	CO4
Unit-V	**EMERGING TRENDS AND INNOVATIONS IN APPLIED MACHINE LEARNING**	**No. of Hours**	**COs**
	Foundation Models and Generative AI Study of large-scale models like GPT, BERT focusing on their architectures, training methodologies, and applications in various industries. Graph Neural Networks (GNNs) Exploration of GNNs and their applications in modeling relational data, such as social networks, recommendation systems, and biological networks. Edge AI and On-device Learning Understanding the deployment of ML models on edge devices, challenges involved, and strategies for efficient on-device learning to enable real-time analytics. AI in Emerging Domains Application of ML techniques in emerging domains such as climate modeling, healthcare diagnostics, and smart agriculture, highlighting innovative solutions and challenges.	9	CO5

Text Books:

1. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron - Covers Unit I & II: Foundational ML concepts, supervised & unsupervised learning, and data preprocessing.
2. Pattern Recognition and Machine Learning" by Christopher M. Bishop - Supports Unit I & IV: Deep dive into mathematical foundations and evaluation techniques.
3. Machine Learning Engineering" by Andriy Burkov - Excellent for Unit III: Practical model deployment, CI/CD, and ML systems in industry
4. Deep Learning with Python" by François Chollet - Addresses Unit V: Includes foundation models, generative AI, and on-device deep learning.

Reference Books:

1. Introduction to Statistical Learning" by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani - Relevant for Units I & II: Statistical methods, linear models, and feature engineering.
2. Designing Machine Learning Systems with Python" by David Julian - Supports Unit III: Implementation, pipelines, and deployment using industry tools
3. Ethics of Artificial Intelligence and Robotics" (Stanford Encyclopedia of Philosophy) - Unit IV: Ethical, societal, and legal implications of ML and AI.
4. Graph Representation Learning" by William L. Hamilton - Unit V: In-depth reference on Graph Neural Networks and their applications.

eLearning Resources:

1. Unit I – Fundamentals of Applied Machine Learning
   Introduction to Machine Learning
   https://onlinecourses.nptel.ac.in/noc23_cs18/preview
2. Unit II – Data Analysis and Machine Learning Techniques for Industrial Applications

	Introduction To Machine Learning - IITKGP https://onlinecourses.nptel.ac.in/noc23_cs87/preview
3.	Unit III – Designing and Implementing ML Solutions with Modern Tools Applied Accelerated Artificial Intelligence https://onlinecourses.nptel.ac.in/noc22_cs83/preview

Note: Number of hours allocated to units includes actual teaching hours and continuous internal assessment hours. Students are given assignments/mini-projects/case-studies/etc as part of experiential learning.